

Supplementary Results for *Fed-MANTIS: Communication-Efficient Federated Adaptation of Time-Series Foundation Models for Wearable Human Activity Recognition*

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Overview

This supplementary document provides the complete experimental result tables for the Fed-MANTIS study. The main manuscript reports the key findings, representative comparisons, and summary tables. This document provides the full centralized and federated result tables for raw-window baselines and MANTIS-based variants on PAMAP2 and MHEALTH.

The supplementary results are organised as follows. Appendix B reports complete centralized raw-window results. Appendix C reports complete federated raw-window results. Appendix D reports complete centralized MANTIS results. Appendix E reports complete Fed-MANTIS results.

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1 Complete Centralized Raw-Window Results

This appendix reports the complete centralized raw-window LOSO results for PAMAP2 and MHEALTH. All models are trained directly on preprocessed sensor windows. The results are reported as mean $\pm$ standard deviation across held-out subjects. The PAMAP2 and MHEALTH results are taken from the final centralized raw-window tuning logs.

1.1 Complete Centralized Raw-Window Results on PAMAP2

Table S1: Complete centralized raw-window LOSO results on PAMAP2.

Model configuration	Model	Optimizer	LR	Accuracy (%)	Macro-F1 (%)
cnn__sgd_lr1e3	CNN	SGD	10^{-3}	85.22 $\pm$ 16.49	78.15 $\pm$ 16.82
cnn__adam_lr1e4	CNN	Adam	10^{-4}	84.21 $\pm$ 18.08	78.63 $\pm$ 19.09
inceptiontime__sgd_lr1e3	InceptionTime	SGD	10^{-3}	83.03 $\pm$ 13.11	74.16 $\pm$ 10.69
cnn__adam_lr1e3	CNN	Adam	10^{-3}	82.89 $\pm$ 17.72	76.42 $\pm$ 19.06
cnn__sgd_lr1e2	CNN	SGD	10^{-2}	82.07 $\pm$ 19.66	74.78 $\pm$ 21.41
inceptiontime__sgd_lr1e2	InceptionTime	SGD	10^{-2}	81.82 $\pm$ 17.84	76.09 $\pm$ 17.56
inceptiontime__adam_lr1e3	InceptionTime	Adam	10^{-3}	81.50 $\pm$ 19.60	73.93 $\pm$ 18.50
inceptiontime__adam_lr1e4	InceptionTime	Adam	10^{-4}	81.44 $\pm$ 17.46	73.77 $\pm$ 16.87
lstm__adam_lr1e3	LSTM	Adam	10^{-3}	79.46 $\pm$ 15.42	71.55 $\pm$ 13.71
lstm__adam_lr1e4	LSTM	Adam	10^{-4}	79.15 $\pm$ 13.37	72.39 $\pm$ 13.76
lstm__sgd_lr1e3	LSTM	SGD	10^{-3}	77.99 $\pm$ 14.69	68.83 $\pm$ 13.35
lstm__sgd_lr1e2	LSTM	SGD	10^{-2}	77.58 $\pm$ 14.31	68.90 $\pm$ 12.98
deepconvlstm__adam_lr1e3	DeepConvLSTM	Adam	10^{-3}	77.17 $\pm$ 19.88	70.59 $\pm$ 20.93
deepconvlstm__sgd_lr1e2	DeepConvLSTM	SGD	10^{-2}	76.03 $\pm$ 19.98	68.90 $\pm$ 18.94
deepconvlstm__sgd_lr1e3	DeepConvLSTM	SGD	10^{-3}	75.81 $\pm$ 20.03	67.17 $\pm$ 18.03
deepconvlstm__adam_lr1e4	DeepConvLSTM	Adam	10^{-4}	75.64 $\pm$ 18.52	65.85 $\pm$ 16.18

The strongest centralized raw-window model on PAMAP2 is CNN with SGD at learning rate 10^{-3} , achieving 85.22% accuracy. The highest macro-F1 among the raw-window configurations is also obtained by a CNN variant, using Adam at learning rate 10^{-4} .

1.2 Complete Centralized Raw-Window Results on MHEALTH

Table S2: Complete centralized raw-window LOSO results on MHEALTH.

Model configuration	Model	Optimizer	LR	Accuracy (%)	Macro-F1 (%)
inceptiontime__adam_lr1e3	InceptionTime	Adam	10^{-3}	94.47 ± 5.90	94.64 ± 5.46
inceptiontime__adam_lr1e4	InceptionTime	Adam	10^{-4}	93.36 ± 5.56	93.42 ± 5.27
inceptiontime__sgd_lr1e2	InceptionTime	SGD	10^{-2}	92.59 ± 5.91	92.65 ± 5.59
inceptiontime__sgd_lr1e3	InceptionTime	SGD	10^{-3}	91.07 ± 5.43	91.27 ± 5.04
lstm__adam_lr1e3	LSTM	Adam	10^{-3}	88.76 ± 8.44	88.52 ± 8.71
deepconvlstm__adam_lr1e3	DeepConvLSTM	Adam	10^{-3}	87.63 ± 6.64	87.80 ± 6.36
deepconvlstm__adam_lr1e4	DeepConvLSTM	Adam	10^{-4}	86.50 ± 7.73	86.95 ± 7.21
cnn__adam_lr1e3	CNN	Adam	10^{-3}	86.40 ± 7.42	86.36 ± 7.70
deepconvlstm__sgd_lr1e2	DeepConvLSTM	SGD	10^{-2}	85.81 ± 9.00	83.71 ± 11.43
lstm__adam_lr1e4	LSTM	Adam	10^{-4}	85.33 ± 6.78	85.51 ± 6.71
cnn__adam_lr1e4	CNN	Adam	10^{-4}	84.87 ± 9.50	84.79 ± 9.95
cnn__sgd_lr1e2	CNN	SGD	10^{-2}	84.59 ± 7.96	84.20 ± 8.29
lstm__sgd_lr1e2	LSTM	SGD	10^{-2}	83.03 ± 11.13	81.98 ± 12.20
cnn__sgd_lr1e3	CNN	SGD	10^{-3}	81.54 ± 9.79	80.76 ± 11.20
deepconvlstm__sgd_lr1e3	DeepConvLSTM	SGD	10^{-3}	78.14 ± 7.74	72.02 ± 7.31
lstm__sgd_lr1e3	LSTM	SGD	10^{-3}	17.99 ± 7.97	9.14 ± 6.54

The strongest centralized raw-window model on MHEALTH is InceptionTime with Adam at learning rate 10^{-3} , achieving 94.47% accuracy and 94.64% macro-F1. This confirms that InceptionTime is the strongest raw-window centralized baseline on MHEALTH, while CNN is the strongest raw-window centralized baseline on PAMAP2.

2 Complete Federated Raw-Window Results

This appendix reports the complete federated raw-window LOSO-FL results for PAMAP2 and MHEALTH. Each model is trained directly on preprocessed sensor windows using weighted FedAvg, where each training subject corresponds to one client. Communication accounting includes both server-to-client download and client-to-server upload model weights. The PAMAP2 and MHEALTH results are taken from the final federated raw-window baseline tuning logs.

2.1 Complete Federated Raw-Window Results on PAMAP2

Table S3: Complete federated raw-window LOSO-FL results on PAMAP2.

Model configuration	Update size	Comm./LOSO	Params	Accuracy (%)	Macro-F1 (%)
cnn__sgd_lr1e2_e1	0.138 MB	57.851 MB	36,108	84.60 ± 15.42	78.84 ± 16.94
cnn__sgd_lr1e2_e3	0.138 MB	57.851 MB	36,108	84.57 ± 14.69	77.41 ± 14.88
cnn__adam_lr1e3_e1	0.138 MB	57.851 MB	36,108	84.38 ± 17.13	77.52 ± 19.34
cnn__sgd_lr1e3_e3	0.138 MB	57.851 MB	36,108	83.77 ± 16.00	76.92 ± 16.21
cnn__adam_lr1e4_e3	0.138 MB	57.851 MB	36,108	83.38 ± 16.26	76.03 ± 18.20
inceptiontime__adam_lr1e3_e1	1.784 MB	749.090 MB	465,740	83.35 ± 14.63	77.71 ± 14.75
inceptiontime__sgd_lr1e3_e1	1.784 MB	749.090 MB	465,740	83.01 ± 12.95	74.67 ± 13.36
cnn__adam_lr1e3_e3	0.138 MB	57.851 MB	36,108	82.82 ± 15.89	73.67 ± 16.57
inceptiontime__adam_lr1e3_e3	1.784 MB	749.090 MB	465,740	82.32 ± 17.03	73.28 ± 16.52
cnn__adam_lr1e4_e1	0.138 MB	57.851 MB	36,108	82.17 ± 13.12	74.10 ± 15.36

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Model configuration	Update size	Comm./LOSO	Params	Accuracy (%)	Macro-F1 (%)
inceptiontime__adam_lr1e4_e3	1.784 MB	749.090 MB	465,740	81.70 $\pm$ 15.51	71.33 $\pm$ 14.70
inceptiontime__sgd_lr1e2_e3	1.784 MB	749.090 MB	465,740	81.46 $\pm$ 13.75	69.74 $\pm$ 11.81
inceptiontime__adam_lr1e4_e1	1.784 MB	749.090 MB	465,740	81.31 $\pm$ 14.24	72.83 $\pm$ 14.66
inceptiontime__sgd_lr1e3_e3	1.784 MB	749.090 MB	465,740	81.06 $\pm$ 14.73	70.54 $\pm$ 12.20
inceptiontime__sgd_lr1e2_e1	1.784 MB	749.090 MB	465,740	79.83 $\pm$ 16.27	72.11 $\pm$ 15.73
deepconvlstm__adam_lr1e3_e3	0.194 MB	81.397 MB	50,540	79.65 $\pm$ 17.69	70.66 $\pm$ 16.21
deepconvlstm__sgd_lr1e2_e3	0.194 MB	81.397 MB	50,540	79.29 $\pm$ 17.47	70.75 $\pm$ 17.98
lstm__adam_lr1e3_e3	2.227 MB	935.175 MB	583,692	79.27 $\pm$ 15.88	69.95 $\pm$ 15.47
lstm__adam_lr1e3_e1	2.227 MB	935.175 MB	583,692	78.33 $\pm$ 18.16	68.43 $\pm$ 16.57
deepconvlstm__adam_lr1e3_e1	0.194 MB	81.397 MB	50,540	77.41 $\pm$ 15.25	69.64 $\pm$ 14.40
deepconvlstm__sgd_lr1e2_e1	0.194 MB	81.397 MB	50,540	77.17 $\pm$ 13.02	65.57 $\pm$ 12.59
deepconvlstm__adam_lr1e4_e3	0.194 MB	81.397 MB	50,540	76.07 $\pm$ 12.37	68.56 $\pm$ 13.15
lstm__adam_lr1e4_e3	2.227 MB	935.175 MB	583,692	74.81 $\pm$ 17.95	64.52 $\pm$ 15.60
lstm__adam_lr1e4_e1	2.227 MB	935.175 MB	583,692	74.36 $\pm$ 18.82	65.00 $\pm$ 18.93
lstm__sgd_lr1e2_e3	2.227 MB	935.175 MB	583,692	73.21 $\pm$ 14.49	63.05 $\pm$ 13.31
deepconvlstm__sgd_lr1e3_e3	0.194 MB	81.397 MB	50,540	71.43 $\pm$ 16.23	63.69 $\pm$ 19.39
deepconvlstm__adam_lr1e4_e1	0.194 MB	81.397 MB	50,540	68.02 $\pm$ 15.31	53.68 $\pm$ 14.67
lstm__sgd_lr1e2_e1	2.227 MB	935.175 MB	583,692	66.93 $\pm$ 15.89	51.54 $\pm$ 14.37
cnn__sgd_lr1e3_e1	0.138 MB	57.851 MB	36,108	65.44 $\pm$ 9.94	51.91 $\pm$ 10.44
lstm__sgd_lr1e3_e3	2.227 MB	935.175 MB	583,692	51.85 $\pm$ 9.07	37.98 $\pm$ 9.45
deepconvlstm__sgd_lr1e3_e1	0.194 MB	81.397 MB	50,540	43.40 $\pm$ 8.24	28.90 $\pm$ 7.19
lstm__sgd_lr1e3_e1	2.227 MB	935.175 MB	583,692	37.07 $\pm$ 8.11	24.37 $\pm$ 9.92

The strongest federated raw-window baseline on PAMAP2 is Fed-CNN with SGD, learning rate 10^{-2} , and one local epoch, achieving 84.60% accuracy and 78.84% macro-F1.

2.2 Complete Federated Raw-Window Results on MHEALTH

Table S4: Complete federated raw-window LOSO-FL results on MHEALTH.

Model configuration	Update size	Comm./LOSO	Params	Accuracy (%)	Macro-F1 (%)
inceptiontime__adam_lr1e3_e1	1.762 MB	951.250 MB	459,980	91.80 $\pm$ 7.19	91.70 $\pm$ 6.94
inceptiontime__adam_lr1e3_e3	1.762 MB	951.250 MB	459,980	91.58 $\pm$ 6.82	91.13 $\pm$ 6.91
inceptiontime__adam_lr1e4_e3	1.762 MB	951.250 MB	459,980	90.32 $\pm$ 8.82	90.18 $\pm$ 8.95
lstm__adam_lr1e3_e3	2.109 MB	1139.087 MB	552,972	89.91 $\pm$ 7.55	89.67 $\pm$ 7.57
inceptiontime__sgd_lr1e2_e3	1.762 MB	951.250 MB	459,980	89.35 $\pm$ 8.21	89.06 $\pm$ 8.25
deepconvlstm__adam_lr1e3_e3	0.175 MB	94.765 MB	45,740	85.50 $\pm$ 6.70	85.18 $\pm$ 7.48
inceptiontime__adam_lr1e4_e1	1.762 MB	951.250 MB	459,980	85.14 $\pm$ 7.18	84.65 $\pm$ 6.85
inceptiontime__sgd_lr1e2_e1	1.762 MB	951.250 MB	459,980	84.15 $\pm$ 6.43	83.66 $\pm$ 6.60
deepconvlstm__adam_lr1e3_e1	0.175 MB	94.765 MB	45,740	83.74 $\pm$ 9.81	83.61 $\pm$ 9.54
lstm__adam_lr1e3_e1	2.109 MB	1139.087 MB	552,972	83.62 $\pm$ 11.11	83.36 $\pm$ 11.19
cnn__adam_lr1e3_e3	0.116 MB	62.515 MB	30,348	83.42 $\pm$ 12.56	83.11 $\pm$ 13.61
cnn__adam_lr1e3_e1	0.116 MB	62.515 MB	30,348	82.61 $\pm$ 11.30	81.90 $\pm$ 12.64
inceptiontime__sgd_lr1e3_e3	1.762 MB	951.250 MB	459,980	80.68 $\pm$ 8.96	75.54 $\pm$ 11.00
cnn__sgd_lr1e2_e3	0.116 MB	62.515 MB	30,348	80.53 $\pm$ 14.10	80.05 $\pm$ 15.51
deepconvlstm__sgd_lr1e2_e3	0.175 MB	94.765 MB	45,740	77.83 $\pm$ 9.94	72.58 $\pm$ 10.71
lstm__adam_lr1e4_e3	2.109 MB	1139.087 MB	552,972	73.53 $\pm$ 10.71	67.38 $\pm$ 10.18
cnn__adam_lr1e4_e3	0.116 MB	62.515 MB	30,348	71.92 $\pm$ 11.60	64.67 $\pm$ 11.22
cnn__sgd_lr1e2_e1	0.116 MB	62.515 MB	30,348	69.25 $\pm$ 11.24	62.68 $\pm$ 11.63
deepconvlstm__adam_lr1e4_e3	0.175 MB	94.765 MB	45,740	68.38 $\pm$ 10.04	60.20 $\pm$ 10.13
inceptiontime__sgd_lr1e3_e1	1.762 MB	951.250 MB	459,980	61.07 $\pm$ 8.45	51.45 $\pm$ 8.65
cnn__sgd_lr1e3_e3	0.116 MB	62.515 MB	30,348	60.75 $\pm$ 8.77	54.01 $\pm$ 8.24
cnn__adam_lr1e4_e1	0.116 MB	62.515 MB	30,348	55.10 $\pm$ 12.92	48.36 $\pm$ 13.49

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Model configuration	Update size	Comm./LOSO	Params	Accuracy (%)	Macro-F1 (%)
deepconvlstm__adam_lr1e4_e1	0.175 MB	94.765 MB	45,740	40.73 $\pm$ 12.48	30.49 $\pm$ 12.01
lstm__adam_lr1e4_e1	2.109 MB	1139.087 MB	552,972	40.60 $\pm$ 15.13	30.77 $\pm$ 14.24
deepconvlstm__sgd_lr1e2_e1	0.175 MB	94.765 MB	45,740	35.84 $\pm$ 7.69	25.77 $\pm$ 7.43
deepconvlstm__sgd_lr1e3_e3	0.175 MB	94.765 MB	45,740	29.23 $\pm$ 7.54	18.54 $\pm$ 6.54
lstm__sgd_lr1e2_e3	2.109 MB	1139.087 MB	552,972	15.80 $\pm$ 4.87	6.23 $\pm$ 3.85
cnn__sgd_lr1e3_e1	0.116 MB	62.515 MB	30,348	12.70 $\pm$ 4.00	7.06 $\pm$ 3.25
lstm__sgd_lr1e3_e3	2.109 MB	1139.087 MB	552,972	9.17 $\pm$ 0.90	1.63 $\pm$ 0.70
deepconvlstm__sgd_lr1e3_e1	0.175 MB	94.765 MB	45,740	8.82 $\pm$ 3.26	2.87 $\pm$ 1.93
lstm__sgd_lr1e3_e1	2.109 MB	1139.087 MB	552,972	8.65 $\pm$ 0.68	1.40 $\pm$ 0.15
lstm__sgd_lr1e2_e1	2.109 MB	1139.087 MB	552,972	8.34 $\pm$ 2.93	1.64 $\pm$ 1.07

The strongest federated raw-window baseline on MHEALTH is Fed-InceptionTime with Adam, learning rate 10^{-3} , and one local epoch, achieving 91.80% accuracy and 91.70% macro-F1.

3 Complete Centralized MANTIS Results

This appendix reports the complete centralized MANTIS-based LOSO results for PAMAP2 and MHEALTH. All models use frozen MANTIS embeddings as input and train only the post-backbone task module. The evaluated variants include the linear probe, low-rank adapters, supervised contrastive low-rank adapters, and confusion-aware low-rank adapters. Results are reported as mean $\pm$ standard deviation across held-out subjects.

3.1 Complete Centralized MANTIS Results on PAMAP2

Table S5: Complete centralized MANTIS LOSO results on PAMAP2.

Variant	Opt.	LR	λ	m	Size	Params	Acc. (%)	Macro-F1 (%)
SupCon LR 256/r64	Adam	10^{-4}	0.05	–	3.263 MB	855,308	94.75 $\pm$ 2.51	92.03 $\pm$ 4.34
Confusion 128/r32	LR Adam	10^{-4}	0.1	0.5	1.616 MB	423,564	94.57 $\pm$ 2.89	93.13 $\pm$ 3.07
Confusion 128/r32	LR Adam	10^{-4}	0.01	0.5	1.616 MB	423,564	94.50 $\pm$ 2.65	90.97 $\pm$ 3.30
Confusion 128/r32	LR Adam	10^{-4}	0.05	1.0	1.616 MB	423,564	94.46 $\pm$ 3.67	91.83 $\pm$ 4.21
Confusion 128/r32	LR Adam	10^{-4}	0.1	1.0	1.616 MB	423,564	94.45 $\pm$ 3.10	92.07 $\pm$ 3.65
SupCon LR 128/r32	Adam	10^{-4}	0.1	–	1.616 MB	423,564	94.40 $\pm$ 3.73	91.79 $\pm$ 4.41
LR adapter 128/r32	Adam	10^{-4}	–	–	1.616 MB	423,564	94.36 $\pm$ 3.21	91.71 $\pm$ 4.19
Confusion 128/r32	LR Adam	10^{-4}	0.01	1.0	1.616 MB	423,564	94.36 $\pm$ 3.42	91.60 $\pm$ 4.64
SupCon LR 256/r64	Adam	10^{-4}	0.1	–	3.263 MB	855,308	94.35 $\pm$ 3.54	93.85 $\pm$ 3.55
Confusion 128/r32	LR Adam	10^{-4}	0.05	0.5	1.616 MB	423,564	94.32 $\pm$ 3.46	92.39 $\pm$ 4.51
SupCon LR 256/r64	Adam	10^{-4}	0.01	–	3.263 MB	855,308	94.27 $\pm$ 3.56	91.43 $\pm$ 4.76
SupCon LR 128/r32	Adam	10^{-4}	0.01	–	1.616 MB	423,564	94.19 $\pm$ 3.24	90.08 $\pm$ 6.15
SupCon LR 128/r32	Adam	10^{-4}	0.05	–	1.616 MB	423,564	94.16 $\pm$ 3.55	90.09 $\pm$ 4.00
LR adapter 256/r64	Adam	10^{-4}	–	–	3.263 MB	855,308	93.96 $\pm$ 3.42	91.35 $\pm$ 4.13
Linear probe	Adam	10^{-4}	–	–	0.598 MB	156,684	93.65 $\pm$ 4.51	91.88 $\pm$ 4.59
SupCon LR 128/r32	SGD	10^{-2}	0.1	–	1.616 MB	423,564	94.61 $\pm$ 3.30	92.80 $\pm$ 4.45
LR adapter 256/r64	SGD	10^{-3}	–	–	3.263 MB	855,308	94.56 $\pm$ 3.02	92.89 $\pm$ 3.37
SupCon LR 256/r64	SGD	10^{-3}	0.05	–	3.263 MB	855,308	94.56 $\pm$ 3.11	92.07 $\pm$ 3.92
SupCon LR 128/r32	SGD	10^{-3}	0.01	–	1.616 MB	423,564	94.55 $\pm$ 3.83	89.45 $\pm$ 4.10
LR adapter 128/r32	SGD	10^{-3}	–	–	1.616 MB	423,564	94.47 $\pm$ 3.49	92.11 $\pm$ 4.05
LR adapter 256/r64	SGD	10^{-2}	–	–	3.263 MB	855,308	94.43 $\pm$ 2.89	91.48 $\pm$ 6.28

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Variant	Opt.	LR	λ	m	Size	Params	Acc. (%)	Macro-F1 (%)
SupCon LR 256/r64	SGD	10^{-2}	0.1	–	3.263 MB	855,308	94.31 ± 3.15	91.67 ± 3.91
SupCon LR 128/r32	SGD	10^{-2}	0.05	–	1.616 MB	423,564	94.26 ± 3.35	91.38 ± 4.28
SupCon LR 256/r64	SGD	10^{-3}	0.1	–	3.263 MB	855,308	94.19 ± 3.29	91.70 ± 4.11
SupCon LR 256/r64	SGD	10^{-3}	0.01	–	3.263 MB	855,308	94.18 ± 3.89	91.67 ± 4.74
Confusion 128/r32	LR SGD	10^{-3}	0.05	1.0	1.616 MB	423,564	94.15 ± 3.72	89.19 ± 3.82
Confusion 128/r32	LR SGD	10^{-2}	0.05	1.0	1.616 MB	423,564	94.05 ± 3.01	91.48 ± 5.48
LR adapter 128/r32	Adam	10^{-3}	–	–	1.616 MB	423,564	94.03 ± 2.52	91.15 ± 6.30
Confusion 128/r32	LR SGD	10^{-2}	0.05	0.5	1.616 MB	423,564	94.01 ± 3.25	90.00 ± 3.85
Confusion 128/r32	LR SGD	10^{-3}	0.05	0.5	1.616 MB	423,564	94.01 ± 3.62	91.35 ± 3.73
SupCon LR 256/r64	SGD	10^{-2}	0.01	–	3.263 MB	855,308	94.00 ± 3.30	92.35 ± 3.96
SupCon LR 256/r64	SGD	10^{-2}	0.05	–	3.263 MB	855,308	93.97 ± 3.51	91.44 ± 4.47
Confusion 128/r32	LR SGD	10^{-3}	0.1	0.5	1.616 MB	423,564	93.97 ± 3.48	90.05 ± 4.19
Linear probe	SGD	10^{-3}	–	–	0.598 MB	156,684	93.96 ± 3.26	88.89 ± 5.74
SupCon LR 256/r64	Adam	10^{-3}	0.1	–	3.263 MB	855,308	93.91 ± 3.53	93.22 ± 3.81
Confusion 128/r32	LR SGD	10^{-3}	0.1	1.0	1.616 MB	423,564	93.87 ± 3.65	92.14 ± 3.98
SupCon LR 128/r32	Adam	10^{-3}	0.1	–	1.616 MB	423,564	93.87 ± 3.59	89.80 ± 4.26
SupCon LR 128/r32	SGD	10^{-3}	0.1	–	1.616 MB	423,564	93.86 ± 3.87	89.91 ± 4.10
SupCon LR 128/r32	SGD	10^{-3}	0.05	–	1.616 MB	423,564	93.74 ± 3.40	88.49 ± 5.77
Confusion 128/r32	LR SGD	10^{-2}	0.1	0.5	1.616 MB	423,564	93.72 ± 3.60	91.04 ± 3.88
Confusion 128/r32	LR SGD	10^{-3}	0.01	0.5	1.616 MB	423,564	93.72 ± 3.59	89.44 ± 3.62
SupCon LR 128/r32	SGD	10^{-2}	0.01	–	1.616 MB	423,564	93.69 ± 3.30	90.70 ± 4.49
SupCon LR 256/r64	Adam	10^{-3}	0.05	–	3.263 MB	855,308	93.64 ± 3.13	89.65 ± 3.85
SupCon LR 256/r64	Adam	10^{-3}	0.01	–	3.263 MB	855,308	93.55 ± 2.47	89.99 ± 3.42
Confusion 128/r32	LR SGD	10^{-3}	0.01	1.0	1.616 MB	423,564	93.54 ± 3.43	89.46 ± 4.39
Confusion 128/r32	LR SGD	10^{-2}	0.1	1.0	1.616 MB	423,564	93.54 ± 4.09	89.41 ± 3.92
Confusion 128/r32	LR Adam	10^{-3}	0.05	0.5	1.616 MB	423,564	93.28 ± 3.40	88.76 ± 5.00
SupCon LR 128/r32	Adam	10^{-3}	0.05	–	1.616 MB	423,564	93.21 ± 4.25	88.27 ± 3.58
LR adapter 256/r64	Adam	10^{-3}	–	–	3.263 MB	855,308	93.15 ± 4.87	89.00 ± 5.08
Confusion 128/r32	LR SGD	10^{-2}	0.01	0.5	1.616 MB	423,564	93.07 ± 4.34	89.22 ± 5.14
Linear probe	SGD	10^{-2}	–	–	0.598 MB	156,684	92.99 ± 3.58	87.72 ± 5.48
Confusion 128/r32	LR Adam	10^{-3}	0.01	1.0	1.616 MB	423,564	92.95 ± 3.04	86.34 ± 9.08
Confusion 128/r32	LR SGD	10^{-2}	0.01	1.0	1.616 MB	423,564	92.84 ± 4.08	89.62 ± 4.41
Confusion 128/r32	LR Adam	10^{-3}	0.1	0.5	1.616 MB	423,564	92.61 ± 5.97	89.26 ± 6.29
Confusion 128/r32	LR Adam	10^{-3}	0.1	1.0	1.616 MB	423,564	92.51 ± 5.07	89.90 ± 5.75
LR adapter 128/r32	SGD	10^{-2}	–	–	1.616 MB	423,564	92.36 ± 4.95	91.45 ± 5.20
Confusion 128/r32	LR Adam	10^{-3}	0.05	1.0	1.616 MB	423,564	92.15 ± 3.79	86.06 ± 7.94
Confusion 128/r32	LR Adam	10^{-3}	0.01	0.5	1.616 MB	423,564	92.01 ± 4.51	88.59 ± 5.93
SupCon LR 128/r32	Adam	10^{-3}	0.01	–	1.616 MB	423,564	91.80 ± 4.62	86.43 ± 5.88
Linear probe	Adam	10^{-3}	–	–	0.598 MB	156,684	91.06 ± 4.31	88.05 ± 6.86

The strongest centralized MANTIS result on PAMAP2 is the SupCon 256/r64 adapter with Adam at learning rate 10^{-4} and $\lambda = 0.05$, achieving 94.75% accuracy. The highest macro-F1 is achieved by the SupCon 256/r64 adapter with $\lambda = 0.1$, reaching 93.85% macro-F1.

3.2 Complete Centralized MANTIS Results on MHEALTH

Table S6: Complete centralized MANTIS LOSO results on MHEALTH.

Variant	Opt.	LR	λ	m	Size	Params	Acc. (%)	Macro-F1 (%)
SupCon LR 256/r64	Adam	10^{-4}	0.05	–	1.388 MB	363,788	97.57 ± 1.99	97.63 ± 2.07
Confusion 128/r32	LR Adam	10^{-4}	0.05	1.0	0.678 MB	177,804	97.53 ± 2.13	97.57 ± 2.27
SupCon LR 128/r32	SGD	10^{-3}	0.05	–	0.678 MB	177,804	97.41 ± 2.20	97.46 ± 2.34
Confusion 128/r32	LR SGD	10^{-3}	0.05	1.0	0.678 MB	177,804	97.29 ± 2.25	97.35 ± 2.37
LR adapter 256/r64	SGD	10^{-2}	–	–	1.388 MB	363,788	97.26 ± 2.56	97.26 ± 2.76
Confusion 128/r32	LR SGD	10^{-3}	0.01	1.0	0.678 MB	177,804	97.26 ± 1.83	97.36 ± 1.88
SupCon LR 128/r32	Adam	10^{-3}	0.1	–	0.678 MB	177,804	97.22 ± 2.59	97.24 ± 2.84
SupCon LR 128/r32	Adam	10^{-4}	0.01	–	0.678 MB	177,804	97.22 ± 2.48	97.21 ± 2.79
Confusion 128/r32	LR SGD	10^{-3}	0.01	0.5	0.678 MB	177,804	97.22 ± 1.97	97.29 ± 2.07
Confusion 128/r32	LR SGD	10^{-3}	0.05	0.5	0.678 MB	177,804	97.19 ± 2.56	97.18 ± 2.86
Confusion 128/r32	LR SGD	10^{-3}	0.1	0.5	0.678 MB	177,804	97.18 ± 2.25	97.26 ± 2.32
Confusion 128/r32	LR Adam	10^{-4}	0.01	1.0	0.678 MB	177,804	97.18 ± 2.26	97.20 ± 2.51
Confusion 128/r32	LR SGD	10^{-2}	0.01	0.5	0.678 MB	177,804	97.13 ± 1.86	97.25 ± 1.88
Confusion 128/r32	LR SGD	10^{-2}	0.1	0.5	0.678 MB	177,804	97.11 ± 2.92	97.02 ± 3.38
Confusion 128/r32	LR Adam	10^{-3}	0.05	0.5	0.678 MB	177,804	97.11 ± 2.84	96.95 ± 3.41
Confusion 128/r32	LR Adam	10^{-4}	0.1	0.5	0.678 MB	177,804	97.10 ± 2.11	97.18 ± 2.20
Confusion 128/r32	LR SGD	10^{-3}	0.1	1.0	0.678 MB	177,804	97.09 ± 1.91	97.22 ± 1.92
SupCon LR 256/r64	SGD	10^{-3}	0.01	–	1.388 MB	363,788	97.08 ± 2.65	97.10 ± 2.81
SupCon LR 256/r64	SGD	10^{-3}	0.1	–	1.388 MB	363,788	97.07 ± 2.26	97.16 ± 2.28
Confusion 128/r32	LR Adam	10^{-4}	0.05	0.5	0.678 MB	177,804	97.06 ± 2.45	96.99 ± 2.92
SupCon LR 256/r64	Adam	10^{-4}	0.1	–	1.388 MB	363,788	97.06 ± 2.44	97.09 ± 2.67
Confusion 128/r32	LR Adam	10^{-4}	0.01	0.5	0.678 MB	177,804	97.06 ± 2.47	97.02 ± 2.85
LR adapter 256/r64	SGD	10^{-3}	–	–	1.388 MB	363,788	97.06 ± 2.09	97.16 ± 2.14
SupCon LR 256/r64	SGD	10^{-3}	0.05	–	1.388 MB	363,788	97.03 ± 2.57	96.91 ± 3.18
LR adapter 128/r32	Adam	10^{-4}	–	–	0.678 MB	177,804	97.02 ± 2.56	97.01 ± 2.83
Confusion 128/r32	LR SGD	10^{-2}	0.01	1.0	0.678 MB	177,804	97.02 ± 2.37	96.97 ± 2.79
SupCon LR 128/r32	SGD	10^{-2}	0.01	–	0.678 MB	177,804	97.00 ± 2.85	96.91 ± 3.31
Confusion 128/r32	LR SGD	10^{-2}	0.1	1.0	0.678 MB	177,804	96.95 ± 2.71	96.89 ± 3.18
SupCon LR 256/r64	SGD	10^{-2}	0.1	–	1.388 MB	363,788	96.92 ± 2.78	96.93 ± 3.06
SupCon LR 256/r64	Adam	10^{-4}	0.01	–	1.388 MB	363,788	96.91 ± 2.75	96.81 ± 3.36
SupCon LR 128/r32	SGD	10^{-3}	0.1	–	0.678 MB	177,804	96.91 ± 2.66	96.84 ± 3.15
SupCon LR 128/r32	Adam	10^{-4}	0.05	–	0.678 MB	177,804	96.91 ± 2.67	96.85 ± 3.13
SupCon LR 256/r64	SGD	10^{-2}	0.01	–	1.388 MB	363,788	96.91 ± 2.56	96.83 ± 3.01
SupCon LR 128/r32	SGD	10^{-2}	0.1	–	0.678 MB	177,804	96.90 ± 2.38	96.95 ± 2.62
SupCon LR 128/r32	Adam	10^{-3}	0.05	–	0.678 MB	177,804	96.90 ± 2.03	97.05 ± 2.03
Confusion 128/r32	LR SGD	10^{-2}	0.05	1.0	0.678 MB	177,804	96.90 ± 2.41	96.88 ± 2.79
SupCon LR 128/r32	SGD	10^{-3}	0.01	–	0.678 MB	177,804	96.87 ± 2.41	96.77 ± 3.04
SupCon LR 256/r64	SGD	10^{-2}	0.05	–	1.388 MB	363,788	96.84 ± 3.00	96.80 ± 3.30
LR adapter 256/r64	Adam	10^{-3}	–	–	1.388 MB	363,788	96.83 ± 2.69	96.78 ± 3.23
SupCon LR 128/r32	Adam	10^{-3}	0.01	–	0.678 MB	177,804	96.82 ± 1.98	96.95 ± 1.97
SupCon LR 128/r32	Adam	10^{-4}	0.1	–	0.678 MB	177,804	96.79 ± 2.46	96.77 ± 2.79

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Variant		Opt.	LR	λ	m	Size	Params	Acc. (%)	Macro-F1 (%)
Linear probe		Adam	10^{-3}	–	–	0.246 MB	64,524	96.64 ± 2.79	96.54 ± 3.32
Confusion	LR	SGD	10^{-2}	0.05	0.5	0.678 MB	177,804	96.64 ± 2.99	96.54 ± 3.58
128/r32									
Confusion	LR	Adam	10^{-3}	0.1	1.0	0.678 MB	177,804	96.62 ± 2.55	96.64 ± 2.86
128/r32									
Confusion	LR	Adam	10^{-4}	0.1	1.0	0.678 MB	177,804	96.57 ± 2.83	96.51 ± 3.29
128/r32									
Linear probe		SGD	10^{-3}	–	–	0.246 MB	64,524	96.57 ± 2.86	96.48 ± 3.39
Linear probe		Adam	10^{-4}	–	–	0.246 MB	64,524	96.57 ± 2.99	96.48 ± 3.44
Confusion	LR	Adam	10^{-3}	0.1	0.5	0.678 MB	177,804	96.54 ± 2.95	96.42 ± 3.42
128/r32									
LR adapter 256/r64		Adam	10^{-4}	–	–	1.388 MB	363,788	96.52 ± 2.76	96.46 ± 3.21
SupCon LR 128/r32		SGD	10^{-2}	0.05	–	0.678 MB	177,804	96.52 ± 2.75	96.55 ± 3.04
LR adapter 128/r32		SGD	10^{-3}	–	–	0.678 MB	177,804	96.48 ± 2.85	96.39 ± 3.29
Confusion	LR	Adam	10^{-3}	0.05	1.0	0.678 MB	177,804	96.45 ± 2.85	96.26 ± 3.49
128/r32									
Confusion	LR	Adam	10^{-3}	0.01	1.0	0.678 MB	177,804	96.32 ± 2.69	96.32 ± 2.99
128/r32									
Confusion	LR	Adam	10^{-3}	0.01	0.5	0.678 MB	177,804	96.19 ± 2.94	96.14 ± 3.33
128/r32									
SupCon LR 256/r64		Adam	10^{-3}	0.1	–	1.388 MB	363,788	95.92 ± 2.85	95.94 ± 3.15
LR adapter 128/r32		Adam	10^{-3}	–	–	0.678 MB	177,804	95.68 ± 2.65	95.60 ± 3.10
SupCon LR 256/r64		Adam	10^{-3}	0.05	–	1.388 MB	363,788	95.65 ± 2.76	95.63 ± 3.12
Linear probe		SGD	10^{-2}	–	–	0.246 MB	64,524	95.11 ± 3.29	94.99 ± 3.74
LR adapter 128/r32		SGD	10^{-2}	–	–	0.678 MB	177,804	95.07 ± 3.32	94.81 ± 3.92
SupCon LR 256/r64		Adam	10^{-3}	0.01	–	1.388 MB	363,788	94.89 ± 4.80	94.80 ± 5.20

The strongest centralized MANTIS result on MHEALTH is the SupCon 256/r64 adapter with Adam at learning rate 10^{-4} and $\lambda = 0.05$, achieving 97.57% accuracy and 97.63% macro-F1.

4 Complete Fed-MANTIS Results

This appendix reports the complete Fed-MANTIS LOSO-FL results for PAMAP2 and MHEALTH. Each training subject is treated as one federated client. All variants use frozen MANTIS embeddings, and only the post-backbone task module is trained and communicated. Communication accounting includes both server-to-client download and client-to-server upload of the task-module parameters.

4.1 Complete Fed-MANTIS Results on PAMAP2

Table S7: Complete Fed-MANTIS LOSO-FL results on PAMAP2.

Variant		Opt.	LR	E	λ	m	Size	Comm./LOSO	Params	Acc. (%)	Macro-F1 (%)
LR	adapter	ADAM	10^{-4}	3	–	–	1.616 MB	678.623 MB	423,564	94.71 ± 3.53	91.95 ± 4.10
128/r32											
Confusion	LR	ADAM	10^{-4}	3	0.1	0.5	1.616 MB	678.623 MB	423,564	94.64 ± 3.19	90.66 ± 4.17
128/r32											
SupCon	LR	ADAM	10^{-4}	3	0.1	–	1.616 MB	678.623 MB	423,564	94.62 ± 3.27	91.98 ± 4.11
128/r32											
Confusion	LR	ADAM	10^{-4}	3	0.1	1.0	1.616 MB	678.623 MB	423,564	94.62 ± 3.23	90.64 ± 4.21
128/r32											
SupCon	LR	ADAM	10^{-4}	1	0.01	–	1.616 MB	678.623 MB	423,564	94.59 ± 3.45	91.86 ± 3.93
128/r32											
Confusion	LR	ADAM	10^{-4}	3	0.05	1.0	1.616 MB	678.623 MB	423,564	94.56 ± 3.39	91.89 ± 4.06
128/r32											

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Variant		Opt.	LR	E	λ	m	Size	Comm./LOSO	Params	Acc. (%)	Macro-F1 (%)
Confusion 128/r32	LR	ADAM	10^{-4}	1	0.05	0.5	1.616 MB	678.623 MB	423,564	94.56 ± 3.37	91.84 ± 3.87
Confusion 128/r32	LR	ADAM	10^{-4}	3	0.05	0.5	1.616 MB	678.623 MB	423,564	94.55 ± 3.38	91.84 ± 4.01
SupCon 128/r32	LR	ADAM	10^{-4}	3	0.05	–	1.616 MB	678.623 MB	423,564	94.54 ± 3.38	91.85 ± 4.02
Confusion 128/r32	LR	ADAM	10^{-4}	1	0.01	0.5	1.616 MB	678.623 MB	423,564	94.53 ± 3.38	92.78 ± 3.62
SupCon 128/r32	LR	ADAM	10^{-4}	1	0.1	–	1.616 MB	678.623 MB	423,564	94.51 ± 3.39	92.78 ± 3.58
Confusion 128/r32	LR	ADAM	10^{-4}	1	0.1	0.5	1.616 MB	678.623 MB	423,564	94.50 ± 3.38	92.73 ± 3.54
SupCon 128/r32	LR	ADAM	10^{-4}	1	0.05	–	1.616 MB	678.623 MB	423,564	94.49 ± 3.36	91.78 ± 3.79
LR 128/r32	adapter	ADAM	10^{-4}	1	–	–	1.616 MB	678.623 MB	423,564	94.49 ± 3.46	92.96 ± 4.07
SupCon 128/r32	LR	ADAM	10^{-4}	3	0.01	–	1.616 MB	678.623 MB	423,564	94.48 ± 3.56	91.76 ± 4.10
SupCon 256/r64	LR	ADAM	10^{-4}	3	0.05	–	3.263 MB	1370.351 MB	855,308	94.42 ± 3.66	91.78 ± 4.01
Confusion 128/r32	LR	ADAM	10^{-4}	3	0.01	1.0	1.616 MB	678.623 MB	423,564	94.42 ± 3.62	91.73 ± 4.13
Confusion 128/r32	LR	ADAM	10^{-4}	3	0.01	0.5	1.616 MB	678.623 MB	423,564	94.42 ± 3.53	91.71 ± 4.10
SupCon 256/r64	LR	ADAM	10^{-4}	3	0.1	–	3.263 MB	1370.351 MB	855,308	94.40 ± 3.58	91.82 ± 3.87
Confusion 128/r32	LR	ADAM	10^{-4}	1	0.1	1.0	1.616 MB	678.623 MB	423,564	94.34 ± 3.36	91.71 ± 3.78
LR 256/r64	adapter	ADAM	10^{-4}	1	–	–	3.263 MB	1370.351 MB	855,308	94.30 ± 4.08	89.28 ± 4.40
Confusion 128/r32	LR	ADAM	10^{-4}	1	0.01	1.0	1.616 MB	678.623 MB	423,564	94.27 ± 3.44	91.61 ± 3.80
SupCon 256/r64	LR	ADAM	10^{-4}	3	0.01	–	3.263 MB	1370.351 MB	855,308	94.26 ± 3.74	91.58 ± 4.08
SupCon 256/r64	LR	ADAM	10^{-4}	1	0.05	–	3.263 MB	1370.351 MB	855,308	94.25 ± 3.91	91.59 ± 4.23
SupCon 256/r64	LR	ADAM	10^{-4}	1	0.01	–	3.263 MB	1370.351 MB	855,308	94.24 ± 4.09	89.24 ± 4.42
Confusion 128/r32	LR	ADAM	10^{-4}	1	0.05	1.0	1.616 MB	678.623 MB	423,564	94.21 ± 3.36	91.56 ± 3.73
SupCon 256/r64	LR	SGD	10^{-3}	3	0.05	–	3.263 MB	1370.351 MB	855,308	94.19 ± 3.94	91.45 ± 4.17
SupCon 256/r64	LR	SGD	10^{-3}	3	0.1	–	3.263 MB	1370.351 MB	855,308	94.15 ± 3.90	91.43 ± 4.12
SupCon 256/r64	LR	SGD	10^{-2}	3	0.1	–	3.263 MB	1370.351 MB	855,308	94.15 ± 4.06	90.50 ± 4.22
SupCon 256/r64	LR	ADAM	10^{-4}	1	0.1	–	3.263 MB	1370.351 MB	855,308	94.14 ± 3.85	92.60 ± 4.34
LR 256/r64	adapter	SGD	10^{-2}	3	–	–	3.263 MB	1370.351 MB	855,308	94.12 ± 3.78	90.44 ± 4.13
SupCon 256/r64	LR	SGD	10^{-3}	3	0.01	–	3.263 MB	1370.351 MB	855,308	94.12 ± 3.98	91.37 ± 4.17
Linear probe LR 256/r64	adapter	ADAM	10^{-4}	3	–	–	0.598 MB	251.035 MB	156,684	94.09 ± 4.57	93.42 ± 4.65
SupCon 128/r32	LR	SGD	10^{-2}	3	0.1	–	1.616 MB	678.623 MB	423,564	94.07 ± 3.36	90.03 ± 3.96
SupCon 128/r32	LR	SGD	10^{-2}	3	0.01	–	1.616 MB	678.623 MB	423,564	94.03 ± 3.47	88.99 ± 3.69
Confusion 128/r32	LR	SGD	10^{-2}	3	0.05	0.5	1.616 MB	678.623 MB	423,564	94.02 ± 3.35	88.95 ± 5.68
Confusion 128/r32	LR	SGD	10^{-3}	3	0.1	0.5	1.616 MB	678.623 MB	423,564	94.02 ± 3.36	88.97 ± 3.71

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Variant		Opt.	LR	E	λ	m	Size	Comm./LOSO	Params	Acc. (%)	Macro-F1 (%)
Confusion 128/r32	LR	SGD	10^{-3}	3	0.1	1.0	1.616 MB	678.623 MB	423,564	94.02 ± 3.37	87.99 ± 5.46
Confusion 128/r32	LR	SGD	10^{-2}	3	0.1	1.0	1.616 MB	678.623 MB	423,564	94.00 ± 3.31	89.98 ± 4.10
SupCon 128/r32	LR	SGD	10^{-3}	3	0.1	–	1.616 MB	678.623 MB	423,564	93.99 ± 3.35	87.89 ± 5.49
Confusion 128/r32	LR	SGD	10^{-2}	3	0.1	0.5	1.616 MB	678.623 MB	423,564	93.99 ± 3.34	88.89 ± 5.74
Confusion 128/r32	LR	SGD	10^{-3}	3	0.05	0.5	1.616 MB	678.623 MB	423,564	93.98 ± 3.35	87.87 ± 5.44
LR 128/r32	adapter	SGD	10^{-2}	3	–	–	1.616 MB	678.623 MB	423,564	93.97 ± 3.65	91.87 ± 4.64
SupCon 256/r64	LR	SGD	10^{-2}	3	0.01	–	3.263 MB	1370.351 MB	855,308	93.97 ± 4.01	92.36 ± 4.50
SupCon 256/r64	LR	SGD	10^{-2}	3	0.05	–	3.263 MB	1370.351 MB	855,308	93.96 ± 3.92	90.36 ± 4.37
SupCon 128/r32	LR	SGD	10^{-2}	3	0.05	–	1.616 MB	678.623 MB	423,564	93.96 ± 3.35	89.91 ± 3.75
Confusion 128/r32	LR	SGD	10^{-3}	3	0.01	0.5	1.616 MB	678.623 MB	423,564	93.95 ± 3.38	87.83 ± 5.43
Confusion 128/r32	LR	SGD	10^{-3}	3	0.05	1.0	1.616 MB	678.623 MB	423,564	93.95 ± 3.37	87.90 ± 5.47
SupCon 128/r32	LR	SGD	10^{-3}	3	0.01	–	1.616 MB	678.623 MB	423,564	93.95 ± 3.39	87.83 ± 5.37
Confusion 128/r32	LR	SGD	10^{-2}	3	0.05	1.0	1.616 MB	678.623 MB	423,564	93.93 ± 3.33	89.95 ± 3.91
SupCon 128/r32	LR	SGD	10^{-3}	3	0.05	–	1.616 MB	678.623 MB	423,564	93.93 ± 3.37	87.80 ± 5.45
LR 128/r32	adapter	SGD	10^{-3}	3	–	–	1.616 MB	678.623 MB	423,564	93.93 ± 3.60	87.77 ± 5.67
LR 256/r64	adapter	ADAM	10^{-4}	3	–	–	3.263 MB	1370.351 MB	855,308	93.88 ± 4.23	91.19 ± 4.39
Confusion 128/r32	LR	SGD	10^{-3}	3	0.01	1.0	1.616 MB	678.623 MB	423,564	93.85 ± 3.52	88.84 ± 3.63
Confusion 128/r32	LR	SGD	10^{-2}	3	0.01	0.5	1.616 MB	678.623 MB	423,564	93.84 ± 3.36	89.81 ± 3.78
Confusion 128/r32	LR	SGD	10^{-2}	3	0.01	1.0	1.616 MB	678.623 MB	423,564	93.74 ± 3.26	88.81 ± 3.57
Linear probe		SGD	10^{-3}	3	–	–	0.598 MB	251.035 MB	156,684	93.36 ± 4.07	89.32 ± 6.43
Linear probe		ADAM	10^{-4}	1	–	–	0.598 MB	251.035 MB	156,684	93.32 ± 4.63	92.66 ± 4.70
Linear probe		SGD	10^{-2}	3	–	–	0.598 MB	251.035 MB	156,684	92.70 ± 4.75	89.40 ± 4.62
SupCon 256/r64	LR	ADAM	10^{-3}	3	0.1	–	3.263 MB	1370.351 MB	855,308	94.60 ± 3.64	91.81 ± 4.83
LR 256/r64	adapter	ADAM	10^{-3}	3	–	–	3.263 MB	1370.351 MB	855,308	94.52 ± 4.27	90.43 ± 4.64
SupCon 256/r64	LR	ADAM	10^{-3}	3	0.05	–	3.263 MB	1370.351 MB	855,308	94.48 ± 3.85	90.83 ± 4.03
Confusion 128/r32	LR	SGD	10^{-2}	1	0.1	1.0	1.616 MB	678.623 MB	423,564	94.45 ± 3.47	91.70 ± 4.11
SupCon 128/r32	LR	SGD	10^{-2}	1	0.1	–	1.616 MB	678.623 MB	423,564	94.45 ± 3.62	90.65 ± 3.83
Confusion 128/r32	LR	SGD	10^{-2}	1	0.1	0.5	1.616 MB	678.623 MB	423,564	94.45 ± 3.60	90.65 ± 3.69
SupCon 128/r32	LR	SGD	10^{-2}	1	0.05	–	1.616 MB	678.623 MB	423,564	94.39 ± 3.53	91.65 ± 3.89
Confusion 128/r32	LR	SGD	10^{-2}	1	0.05	0.5	1.616 MB	678.623 MB	423,564	94.39 ± 3.52	91.68 ± 3.89
Confusion 128/r32	LR	SGD	10^{-2}	1	0.05	1.0	1.616 MB	678.623 MB	423,564	94.29 ± 3.57	91.55 ± 4.01
LR 128/r32	adapter	SGD	10^{-3}	1	–	–	1.616 MB	678.623 MB	423,564	94.25 ± 3.54	92.22 ± 4.55
SupCon 128/r32	LR	SGD	10^{-2}	1	0.01	–	1.616 MB	678.623 MB	423,564	94.24 ± 3.61	91.44 ± 4.12

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Variant		Opt.	LR	E	λ	m	Size	Comm./LOSO	Params	Acc. (%)	Macro-F1 (%)
Confusion 128/r32	LR	SGD	10^{-2}	1	0.01	1.0	1.616 MB	678.623 MB	423,564	94.21 ± 3.56	90.14 ± 4.19
SupCon 256/r64	LR	ADAM	10^{-3}	3	0.01	–	3.263 MB	1370.351 MB	855,308	94.19 ± 3.79	90.41 ± 5.38
Confusion 128/r32	LR	SGD	10^{-2}	1	0.01	0.5	1.616 MB	678.623 MB	423,564	94.17 ± 3.54	91.37 ± 4.00
Linear probe		ADAM	10^{-3}	3	–	–	0.598 MB	251.035 MB	156,684	94.10 ± 4.11	92.06 ± 4.87
SupCon 128/r32	LR	ADAM	10^{-3}	3	0.05	–	1.616 MB	678.623 MB	423,564	94.10 ± 3.73	89.20 ± 5.10
LR 128/r32	adapter	SGD	10^{-2}	1	–	–	1.616 MB	678.623 MB	423,564	94.09 ± 3.51	91.07 ± 5.06
Confusion 128/r32	LR	ADAM	10^{-3}	3	0.05	1.0	1.616 MB	678.623 MB	423,564	94.07 ± 3.81	91.13 ± 4.28
LR 256/r64	adapter	SGD	10^{-2}	1	–	–	3.263 MB	1370.351 MB	855,308	94.06 ± 3.56	91.05 ± 4.64
Confusion 128/r32	LR	ADAM	10^{-3}	1	0.1	1.0	1.616 MB	678.623 MB	423,564	94.05 ± 3.47	91.38 ± 3.88
Confusion 128/r32	LR	ADAM	10^{-3}	3	0.05	0.5	1.616 MB	678.623 MB	423,564	94.05 ± 3.70	91.25 ± 3.90
Confusion 128/r32	LR	ADAM	10^{-3}	3	0.1	0.5	1.616 MB	678.623 MB	423,564	94.03 ± 3.78	92.51 ± 4.34
Confusion 128/r32	LR	ADAM	10^{-3}	3	0.1	1.0	1.616 MB	678.623 MB	423,564	93.99 ± 3.94	89.97 ± 4.10
LR 256/r64	adapter	SGD	10^{-3}	1	–	–	3.263 MB	1370.351 MB	855,308	93.98 ± 3.93	90.08 ± 6.70
SupCon 128/r32	LR	ADAM	10^{-3}	1	0.05	–	1.616 MB	678.623 MB	423,564	93.97 ± 3.82	90.93 ± 4.74
Confusion 128/r32	LR	ADAM	10^{-3}	3	0.01	0.5	1.616 MB	678.623 MB	423,564	93.97 ± 4.14	89.85 ± 4.32
SupCon 256/r64	LR	SGD	10^{-3}	1	0.01	–	3.263 MB	1370.351 MB	855,308	93.94 ± 3.93	90.05 ± 6.67
SupCon 256/r64	LR	SGD	10^{-3}	1	0.05	–	3.263 MB	1370.351 MB	855,308	93.91 ± 3.93	90.02 ± 6.61
SupCon 128/r32	LR	ADAM	10^{-3}	1	0.01	–	1.616 MB	678.623 MB	423,564	93.91 ± 4.06	90.83 ± 4.64
SupCon 256/r64	LR	SGD	10^{-3}	1	0.1	–	3.263 MB	1370.351 MB	855,308	93.90 ± 3.94	90.00 ± 6.58
Confusion 128/r32	LR	ADAM	10^{-3}	1	0.01	0.5	1.616 MB	678.623 MB	423,564	93.87 ± 3.92	91.23 ± 4.33
SupCon 256/r64	LR	SGD	10^{-2}	1	0.05	–	3.263 MB	1370.351 MB	855,308	93.85 ± 3.71	89.82 ± 4.18
SupCon 256/r64	LR	SGD	10^{-2}	1	0.01	–	3.263 MB	1370.351 MB	855,308	93.85 ± 3.58	90.85 ± 4.61
SupCon 128/r32	LR	ADAM	10^{-3}	1	0.1	–	1.616 MB	678.623 MB	423,564	93.85 ± 3.53	91.03 ± 3.69
Confusion 128/r32	LR	ADAM	10^{-3}	1	0.1	0.5	1.616 MB	678.623 MB	423,564	93.83 ± 3.51	91.79 ± 4.36
Confusion 128/r32	LR	ADAM	10^{-3}	3	0.01	1.0	1.616 MB	678.623 MB	423,564	93.83 ± 3.62	89.83 ± 4.31
LR 256/r64	adapter	ADAM	10^{-3}	1	–	–	3.263 MB	1370.351 MB	855,308	93.80 ± 3.88	90.87 ± 6.76
SupCon 256/r64	LR	SGD	10^{-2}	1	0.1	–	3.263 MB	1370.351 MB	855,308	93.78 ± 3.90	89.79 ± 4.21
Linear probe		ADAM	10^{-3}	1	–	–	0.598 MB	251.035 MB	156,684	93.75 ± 4.25	93.12 ± 4.38
SupCon 128/r32	LR	ADAM	10^{-3}	3	0.1	–	1.616 MB	678.623 MB	423,564	93.70 ± 3.50	90.12 ± 5.20
Confusion 128/r32	LR	SGD	10^{-3}	1	0.1	1.0	1.616 MB	678.623 MB	423,564	93.70 ± 3.48	91.62 ± 4.52
SupCon 128/r32	LR	SGD	10^{-3}	1	0.1	–	1.616 MB	678.623 MB	423,564	93.66 ± 3.68	92.83 ± 4.05
Confusion 128/r32	LR	SGD	10^{-3}	1	0.1	0.5	1.616 MB	678.623 MB	423,564	93.66 ± 3.61	92.83 ± 3.94
Confusion 128/r32	LR	SGD	10^{-3}	1	0.05	1.0	1.616 MB	678.623 MB	423,564	93.62 ± 3.55	92.85 ± 3.85

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Variant		Opt.	LR	E	λ	m	Size	Comm./LOSO	Params	Acc. (%)	Macro-F1 (%)
SupCon 128/r32	LR	ADAM	10^{-3}	3	0.01	–	1.616 MB	678.623 MB	423,564	93.61 ± 3.75	90.83 ± 4.04
SupCon 256/r64	LR	ADAM	10^{-3}	1	0.1	–	3.263 MB	1370.351 MB	855,308	93.59 ± 3.77	87.53 ± 6.52
Confusion 128/r32	LR	SGD	10^{-3}	1	0.05	0.5	1.616 MB	678.623 MB	423,564	93.56 ± 3.62	92.77 ± 3.98
SupCon 128/r32	LR	SGD	10^{-3}	1	0.01	–	1.616 MB	678.623 MB	423,564	93.56 ± 3.67	92.81 ± 4.06
SupCon 128/r32	LR	SGD	10^{-3}	1	0.05	–	1.616 MB	678.623 MB	423,564	93.55 ± 3.62	92.79 ± 3.97
Confusion 128/r32	LR	ADAM	10^{-3}	1	0.05	0.5	1.616 MB	678.623 MB	423,564	93.53 ± 4.02	91.89 ± 4.82
Confusion 128/r32	LR	SGD	10^{-3}	1	0.01	0.5	1.616 MB	678.623 MB	423,564	93.50 ± 3.60	92.73 ± 3.94
Confusion 128/r32	LR	SGD	10^{-3}	1	0.01	1.0	1.616 MB	678.623 MB	423,564	93.50 ± 3.62	92.71 ± 3.98
Confusion 128/r32	LR	ADAM	10^{-3}	1	0.05	1.0	1.616 MB	678.623 MB	423,564	93.47 ± 3.72	91.50 ± 3.92
LR 128/r32	adapter	ADAM	10^{-3}	3	–	–	1.616 MB	678.623 MB	423,564	93.46 ± 3.72	89.39 ± 6.24
Confusion 128/r32	LR	ADAM	10^{-3}	1	0.01	1.0	1.616 MB	678.623 MB	423,564	93.38 ± 4.09	91.68 ± 4.60
LR 128/r32	adapter	ADAM	10^{-3}	1	–	–	1.616 MB	678.623 MB	423,564	93.34 ± 4.10	87.53 ± 7.91
SupCon 256/r64	LR	ADAM	10^{-3}	1	0.01	–	3.263 MB	1370.351 MB	855,308	93.31 ± 4.22	90.42 ± 5.25
SupCon 256/r64	LR	ADAM	10^{-3}	1	0.05	–	3.263 MB	1370.351 MB	855,308	93.21 ± 4.38	90.18 ± 5.07
Linear probe		SGD	10^{-3}	1	–	–	0.598 MB	251.035 MB	156,684	92.96 ± 3.87	88.95 ± 6.49
Linear probe		SGD	10^{-2}	1	–	–	0.598 MB	251.035 MB	156,684	92.42 ± 4.36	86.23 ± 5.58

4.2 Complete Fed-MANTIS Results on MHEALTH

Table S8: Complete Fed-MANTIS LOSO-FL results on MHEALTH.

Variant		Opt.	LR	E	λ	m	Size	Comm./LOSO	Params	Acc. (%)	Macro-F1 (%)
SupCon 256/r64	LR	SGD	10^{-2}	3	0.1	–	1.388 MB	749.380 MB	363,788	97.38 ± 2.66	97.33 ± 3.05
LR 256/r64	adapter	ADAM	10^{-4}	3	–	–	1.388 MB	749.380 MB	363,788	97.29 ± 2.08	97.36 ± 2.18
SupCon 128/r32	LR	ADAM	10^{-3}	3	0.1	–	0.678 MB	366.265 MB	177,804	97.22 ± 2.41	97.31 ± 2.44
LR 256/r64	adapter	SGD	10^{-2}	3	–	–	1.388 MB	749.380 MB	363,788	97.22 ± 2.49	97.18 ± 2.88
SupCon 128/r32	LR	ADAM	10^{-3}	3	0.05	–	0.678 MB	366.265 MB	177,804	97.22 ± 2.23	97.31 ± 2.27
SupCon 256/r64	LR	SGD	10^{-2}	3	0.05	–	1.388 MB	749.380 MB	363,788	97.18 ± 2.60	97.14 ± 2.98
Confusion 128/r32	LR	ADAM	10^{-3}	3	0.1	0.5	0.678 MB	366.265 MB	177,804	97.10 ± 2.28	97.21 ± 2.30
SupCon 128/r32	LR	ADAM	10^{-3}	3	0.01	–	0.678 MB	366.265 MB	177,804	97.10 ± 2.23	97.21 ± 2.23
Confusion 128/r32	LR	ADAM	10^{-3}	3	0.1	1.0	0.678 MB	366.265 MB	177,804	97.05 ± 2.01	97.18 ± 1.99
SupCon 256/r64	LR	ADAM	10^{-4}	3	0.05	–	1.388 MB	749.380 MB	363,788	97.03 ± 2.68	96.99 ± 3.07
SupCon 256/r64	LR	ADAM	10^{-4}	3	0.1	–	1.388 MB	749.380 MB	363,788	97.03 ± 2.67	96.98 ± 3.04
Confusion 128/r32	LR	ADAM	10^{-3}	3	0.05	0.5	0.678 MB	366.265 MB	177,804	97.02 ± 2.74	97.04 ± 3.03

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Variant		Opt.	LR	E	λ	m	Size	Comm./LOSO	Params	Acc. (%)	Macro-F1 (%)
LR 128/r32	adapter	ADAM	10^{-4}	3	–	–	0.678 MB	366.265 MB	177,804	96.99 ± 2.62	96.99 ± 2.92
SupCon 256/r64	LR	ADAM	10^{-3}	3	0.1	–	1.388 MB	749.380 MB	363,788	96.98 ± 2.40	96.97 ± 2.69
SupCon 256/r64	LR	SGD	10^{-2}	3	0.01	–	1.388 MB	749.380 MB	363,788	96.95 ± 2.67	96.93 ± 3.06
SupCon 256/r64	LR	ADAM	10^{-4}	3	0.01	–	1.388 MB	749.380 MB	363,788	96.93 ± 2.50	96.95 ± 2.79
SupCon 256/r64	LR	ADAM	10^{-3}	3	0.05	–	1.388 MB	749.380 MB	363,788	96.91 ± 2.79	96.81 ± 3.24
Confusion 128/r32	LR	ADAM	10^{-3}	3	0.05	1.0	0.678 MB	366.265 MB	177,804	96.86 ± 2.22	96.99 ± 2.22
SupCon 256/r64	LR	ADAM	10^{-3}	3	0.01	–	1.388 MB	749.380 MB	363,788	96.78 ± 2.60	96.72 ± 3.01
Confusion 128/r32	LR	SGD	10^{-2}	3	0.1	1.0	0.678 MB	366.265 MB	177,804	96.76 ± 2.59	96.74 ± 2.95
SupCon 256/r64	LR	ADAM	10^{-3}	1	0.05	–	1.388 MB	749.380 MB	363,788	96.74 ± 2.56	96.69 ± 3.02
Linear probe		SGD	10^{-3}	3	–	–	0.246 MB	132.915 MB	64,524	96.71 ± 2.44	96.70 ± 2.78
Linear probe		ADAM	10^{-3}	3	–	–	0.246 MB	132.915 MB	64,524	96.70 ± 2.65	96.62 ± 3.20
Confusion 128/r32	LR	SGD	10^{-2}	3	0.1	0.5	0.678 MB	366.265 MB	177,804	96.68 ± 2.59	96.66 ± 2.95
SupCon 128/r32	LR	SGD	10^{-2}	3	0.1	–	0.678 MB	366.265 MB	177,804	96.64 ± 2.49	96.62 ± 2.87
Confusion 128/r32	LR	SGD	10^{-2}	3	0.05	1.0	0.678 MB	366.265 MB	177,804	96.61 ± 2.84	96.58 ± 3.17
LR 256/r64	adapter	ADAM	10^{-3}	3	–	–	1.388 MB	749.380 MB	363,788	96.60 ± 2.72	96.63 ± 2.85
LR 256/r64	adapter	ADAM	10^{-3}	1	–	–	1.388 MB	749.380 MB	363,788	96.59 ± 1.14	96.77 ± 1.12
SupCon 128/r32	LR	SGD	10^{-2}	3	0.01	–	0.678 MB	366.265 MB	177,804	96.57 ± 2.62	96.53 ± 2.98
SupCon 256/r64	LR	SGD	10^{-3}	3	0.1	–	1.388 MB	749.380 MB	363,788	96.55 ± 2.57	96.61 ± 2.81
LR 128/r32	adapter	ADAM	10^{-3}	3	–	–	0.678 MB	366.265 MB	177,804	96.55 ± 2.94	96.59 ± 3.03
SupCon 256/r64	LR	ADAM	10^{-3}	1	0.1	–	1.388 MB	749.380 MB	363,788	96.53 ± 3.07	96.45 ± 3.45
Confusion 128/r32	LR	SGD	10^{-2}	3	0.01	1.0	0.678 MB	366.265 MB	177,804	96.53 ± 2.88	96.49 ± 3.20
Confusion 128/r32	LR	SGD	10^{-2}	3	0.05	0.5	0.678 MB	366.265 MB	177,804	96.53 ± 2.74	96.49 ± 3.08
SupCon 128/r32	LR	SGD	10^{-2}	3	0.05	–	0.678 MB	366.265 MB	177,804	96.53 ± 2.60	96.50 ± 2.96
LR 128/r32	adapter	SGD	10^{-2}	3	–	–	0.678 MB	366.265 MB	177,804	96.52 ± 2.57	96.43 ± 3.17
Confusion 128/r32	LR	SGD	10^{-2}	3	0.01	0.5	0.678 MB	366.265 MB	177,804	96.45 ± 2.76	96.43 ± 3.09
SupCon 128/r32	LR	ADAM	10^{-4}	3	0.01	–	0.678 MB	366.265 MB	177,804	96.44 ± 2.67	96.46 ± 2.97
SupCon 256/r64	LR	SGD	10^{-3}	3	0.01	–	1.388 MB	749.380 MB	363,788	96.43 ± 2.60	96.50 ± 2.79
LR 256/r64	adapter	SGD	10^{-3}	3	–	–	1.388 MB	749.380 MB	363,788	96.39 ± 2.65	96.45 ± 2.86
SupCon 256/r64	LR	SGD	10^{-2}	1	0.05	–	1.388 MB	749.380 MB	363,788	96.39 ± 2.34	96.42 ± 2.66
SupCon 256/r64	LR	SGD	10^{-3}	3	0.05	–	1.388 MB	749.380 MB	363,788	96.39 ± 2.74	96.47 ± 2.89
SupCon 256/r64	LR	SGD	10^{-2}	1	0.1	–	1.388 MB	749.380 MB	363,788	96.38 ± 2.17	96.45 ± 2.43
Confusion 128/r32	LR	ADAM	10^{-3}	3	0.01	1.0	0.678 MB	366.265 MB	177,804	96.35 ± 2.61	96.43 ± 2.67
Confusion 128/r32	LR	ADAM	10^{-3}	1	0.05	1.0	0.678 MB	366.265 MB	177,804	96.35 ± 2.06	96.50 ± 2.14

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Variant		Opt.	LR	E	λ	m	Size	Comm./LOSO	Params	Acc. (%)	Macro-F1 (%)
SupCon 256/r64	LR	ADAM	10^{-4}	1	0.01	–	1.388 MB	749.380 MB	363,788	96.32 ± 2.47	96.38 ± 2.70
Confusion 128/r32	LR	ADAM	10^{-4}	3	0.05	1.0	0.678 MB	366.265 MB	177,804	96.29 ± 2.78	96.33 ± 2.98
LR 256/r64	adapter	SGD	10^{-2}	1	–	–	1.388 MB	749.380 MB	363,788	96.28 ± 2.56	96.32 ± 2.87
Confusion 128/r32	LR	ADAM	10^{-4}	3	0.01	1.0	0.678 MB	366.265 MB	177,804	96.27 ± 2.56	96.31 ± 2.82
SupCon 128/r32	LR	ADAM	10^{-4}	3	0.05	–	0.678 MB	366.265 MB	177,804	96.26 ± 2.86	96.29 ± 3.06
Confusion 128/r32	LR	ADAM	10^{-4}	3	0.05	0.5	0.678 MB	366.265 MB	177,804	96.25 ± 2.77	96.29 ± 2.97
LR 256/r64	adapter	ADAM	10^{-4}	1	–	–	1.388 MB	749.380 MB	363,788	96.24 ± 2.62	96.24 ± 3.00
SupCon 256/r64	LR	SGD	10^{-2}	1	0.01	–	1.388 MB	749.380 MB	363,788	96.20 ± 2.52	96.24 ± 2.85
Linear probe		SGD	10^{-2}	3	–	–	0.246 MB	132.915 MB	64,524	96.20 ± 3.03	96.21 ± 3.34
Confusion 128/r32	LR	ADAM	10^{-3}	3	0.01	0.5	0.678 MB	366.265 MB	177,804	96.18 ± 2.92	96.14 ± 3.24
Confusion 128/r32	LR	ADAM	10^{-4}	3	0.01	0.5	0.678 MB	366.265 MB	177,804	96.11 ± 2.46	96.16 ± 2.73
SupCon 256/r64	LR	ADAM	10^{-3}	1	0.01	–	1.388 MB	749.380 MB	363,788	96.10 ± 3.10	96.07 ± 3.36
Confusion 128/r32	LR	ADAM	10^{-3}	1	0.01	0.5	0.678 MB	366.265 MB	177,804	96.09 ± 3.03	96.23 ± 3.17
Confusion 128/r32	LR	ADAM	10^{-4}	3	0.1	1.0	0.678 MB	366.265 MB	177,804	96.09 ± 3.06	96.11 ± 3.21
SupCon 256/r64	LR	ADAM	10^{-4}	1	0.05	–	1.388 MB	749.380 MB	363,788	96.07 ± 2.51	96.16 ± 2.67
Linear probe		SGD	10^{-2}	1	–	–	0.246 MB	132.915 MB	64,524	96.05 ± 2.84	96.02 ± 3.24
LR 128/r32	adapter	SGD	10^{-3}	3	–	–	0.678 MB	366.265 MB	177,804	96.02 ± 3.31	96.09 ± 3.57
SupCon 256/r64	LR	ADAM	10^{-4}	1	0.1	–	1.388 MB	749.380 MB	363,788	96.02 ± 2.53	96.11 ± 2.64
Confusion 128/r32	LR	ADAM	10^{-4}	3	0.1	0.5	0.678 MB	366.265 MB	177,804	96.01 ± 2.99	96.04 ± 3.15
SupCon 128/r32	LR	ADAM	10^{-4}	3	0.1	–	0.678 MB	366.265 MB	177,804	95.88 ± 3.23	95.87 ± 3.41
Linear probe		ADAM	10^{-4}	3	–	–	0.246 MB	132.915 MB	64,524	95.81 ± 3.86	95.60 ± 4.31
SupCon 128/r32	LR	ADAM	10^{-3}	1	0.05	–	0.678 MB	366.265 MB	177,804	95.77 ± 3.09	95.95 ± 3.08
SupCon 128/r32	LR	ADAM	10^{-3}	1	0.1	–	0.678 MB	366.265 MB	177,804	95.68 ± 3.57	95.79 ± 3.61
Confusion 128/r32	LR	ADAM	10^{-3}	1	0.01	1.0	0.678 MB	366.265 MB	177,804	95.64 ± 3.51	95.69 ± 3.62
Linear probe		SGD	10^{-3}	1	–	–	0.246 MB	132.915 MB	64,524	95.41 ± 2.59	95.52 ± 2.72
Confusion 128/r32	LR	ADAM	10^{-3}	1	0.05	0.5	0.678 MB	366.265 MB	177,804	95.41 ± 3.93	95.40 ± 4.28
LR 128/r32	adapter	ADAM	10^{-3}	1	–	–	0.678 MB	366.265 MB	177,804	95.30 ± 4.29	95.13 ± 4.89
Confusion 128/r32	LR	ADAM	10^{-3}	1	0.1	0.5	0.678 MB	366.265 MB	177,804	95.26 ± 3.80	95.12 ± 4.42
Confusion 128/r32	LR	ADAM	10^{-3}	1	0.1	1.0	0.678 MB	366.265 MB	177,804	95.22 ± 3.86	95.09 ± 4.46
SupCon 128/r32	LR	ADAM	10^{-3}	1	0.01	–	0.678 MB	366.265 MB	177,804	95.13 ± 2.57	95.28 ± 2.62
LR 128/r32	adapter	SGD	10^{-2}	1	–	–	0.678 MB	366.265 MB	177,804	95.08 ± 4.33	95.04 ± 4.85
Linear probe		ADAM	10^{-3}	1	–	–	0.246 MB	132.915 MB	64,524	94.89 ± 3.75	94.70 ± 4.13
Linear probe		ADAM	10^{-4}	1	–	–	0.246 MB	132.915 MB	64,524	94.41 ± 3.34	94.38 ± 3.68
Confusion 128/r32	LR	SGD	10^{-3}	3	0.05	0.5	0.678 MB	366.265 MB	177,804	94.07 ± 4.24	94.17 ± 4.39
Confusion 128/r32	LR	SGD	10^{-3}	3	0.05	1.0	0.678 MB	366.265 MB	177,804	93.94 ± 4.50	94.08 ± 4.54

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Variant		Opt.	LR	E	λ	m	Size	Comm./LOSO	Params	Acc. (%)	Macro-F1 (%)
LR 128/r32	adapter	ADAM	10^{-4}	1	–	–	0.678 MB	366.265 MB	177,804	93.84 ± 4.44	93.73 ± 4.79
SupCon 128/r32	LR	SGD	10^{-3}	3	0.05	–	0.678 MB	366.265 MB	177,804	93.79 ± 4.76	93.93 ± 4.73
Confusion 128/r32	LR	SGD	10^{-3}	3	0.01	0.5	0.678 MB	366.265 MB	177,804	93.74 ± 4.75	93.91 ± 4.75
SupCon 128/r32	LR	SGD	10^{-3}	3	0.01	–	0.678 MB	366.265 MB	177,804	93.66 ± 4.89	93.84 ± 4.86
Confusion 128/r32	LR	SGD	10^{-3}	3	0.1	0.5	0.678 MB	366.265 MB	177,804	93.62 ± 4.89	93.83 ± 4.78
Confusion 128/r32	LR	SGD	10^{-3}	3	0.1	1.0	0.678 MB	366.265 MB	177,804	93.58 ± 4.91	93.80 ± 4.79
Confusion 128/r32	LR	SGD	10^{-3}	3	0.01	1.0	0.678 MB	366.265 MB	177,804	93.54 ± 4.95	93.72 ± 4.90
SupCon 128/r32	LR	SGD	10^{-3}	3	0.1	–	0.678 MB	366.265 MB	177,804	93.47 ± 4.89	93.67 ± 4.77
Confusion 128/r32	LR	ADAM	10^{-4}	1	0.01	0.5	0.678 MB	366.265 MB	177,804	93.43 ± 4.02	93.35 ± 4.34
Confusion 128/r32	LR	ADAM	10^{-4}	1	0.01	1.0	0.678 MB	366.265 MB	177,804	93.39 ± 3.99	93.31 ± 4.31
Confusion 128/r32	LR	ADAM	10^{-4}	1	0.05	1.0	0.678 MB	366.265 MB	177,804	93.19 ± 4.19	93.08 ± 4.54
Confusion 128/r32	LR	ADAM	10^{-4}	1	0.05	0.5	0.678 MB	366.265 MB	177,804	93.19 ± 4.20	93.08 ± 4.55
SupCon 128/r32	LR	ADAM	10^{-4}	1	0.01	–	0.678 MB	366.265 MB	177,804	93.18 ± 4.12	93.13 ± 4.40
SupCon 128/r32	LR	ADAM	10^{-4}	1	0.05	–	0.678 MB	366.265 MB	177,804	93.03 ± 4.36	92.93 ± 4.70
Confusion 128/r32	LR	ADAM	10^{-4}	1	0.1	0.5	0.678 MB	366.265 MB	177,804	92.95 ± 4.46	92.87 ± 4.75
Confusion 128/r32	LR	ADAM	10^{-4}	1	0.1	1.0	0.678 MB	366.265 MB	177,804	92.95 ± 4.46	92.87 ± 4.75
SupCon 128/r32	LR	ADAM	10^{-4}	1	0.1	–	0.678 MB	366.265 MB	177,804	92.63 ± 4.39	92.57 ± 4.68
Confusion 128/r32	LR	SGD	10^{-2}	1	0.1	0.5	0.678 MB	366.265 MB	177,804	91.99 ± 5.12	91.40 ± 5.60
Confusion 128/r32	LR	SGD	10^{-2}	1	0.1	1.0	0.678 MB	366.265 MB	177,804	91.76 ± 4.81	91.13 ± 5.29
Confusion 128/r32	LR	SGD	10^{-2}	1	0.05	1.0	0.678 MB	366.265 MB	177,804	91.57 ± 4.78	90.99 ± 5.18
SupCon 128/r32	LR	SGD	10^{-2}	1	0.1	–	0.678 MB	366.265 MB	177,804	91.57 ± 5.06	90.96 ± 5.48
Confusion 128/r32	LR	SGD	10^{-2}	1	0.01	1.0	0.678 MB	366.265 MB	177,804	91.52 ± 4.70	90.85 ± 5.07
Confusion 128/r32	LR	SGD	10^{-2}	1	0.05	0.5	0.678 MB	366.265 MB	177,804	91.49 ± 4.89	90.93 ± 5.28
Confusion 128/r32	LR	SGD	10^{-2}	1	0.01	0.5	0.678 MB	366.265 MB	177,804	91.49 ± 4.61	90.82 ± 5.00
SupCon 128/r32	LR	SGD	10^{-2}	1	0.05	–	0.678 MB	366.265 MB	177,804	91.45 ± 5.19	90.84 ± 5.54
SupCon 128/r32	LR	SGD	10^{-2}	1	0.01	–	0.678 MB	366.265 MB	177,804	91.33 ± 4.88	90.68 ± 5.23
SupCon 256/r64	LR	SGD	10^{-3}	1	0.01	–	1.388 MB	749.380 MB	363,788	90.35 ± 4.86	89.07 ± 5.51
LR 256/r64	adapter	SGD	10^{-3}	1	–	–	1.388 MB	749.380 MB	363,788	90.35 ± 4.94	89.11 ± 5.55
SupCon 256/r64	LR	SGD	10^{-3}	1	0.1	–	1.388 MB	749.380 MB	363,788	90.27 ± 4.34	89.31 ± 4.90
SupCon 256/r64	LR	SGD	10^{-3}	1	0.05	–	1.388 MB	749.380 MB	363,788	90.24 ± 4.98	88.88 ± 5.50
LR 128/r32	adapter	SGD	10^{-3}	1	–	–	0.678 MB	366.265 MB	177,804	87.80 ± 6.36	87.43 ± 7.03
Confusion 128/r32	LR	SGD	10^{-3}	1	0.05	0.5	0.678 MB	366.265 MB	177,804	79.30 ± 8.18	72.86 ± 8.21

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Variant		Opt.	LR	E	λ	m	Size	Comm./LOSO	Params	Acc. (%)	Macro-F1 (%)
Confusion 128/r32	LR	SGD	10^{-3}	1	0.05	1.0	0.678 MB	366.265 MB	177,804	79.30 ± 8.18	72.86 ± 8.21
Confusion 128/r32	LR	SGD	10^{-3}	1	0.1	1.0	0.678 MB	366.265 MB	177,804	78.93 ± 8.55	72.23 ± 9.19
Confusion 128/r32	LR	SGD	10^{-3}	1	0.01	0.5	0.678 MB	366.265 MB	177,804	78.89 ± 7.41	72.86 ± 7.04
Confusion 128/r32	LR	SGD	10^{-3}	1	0.01	1.0	0.678 MB	366.265 MB	177,804	78.85 ± 7.37	72.83 ± 7.02
Confusion 128/r32	LR	SGD	10^{-3}	1	0.1	0.5	0.678 MB	366.265 MB	177,804	78.85 ± 8.59	72.10 ± 9.25
SupCon 128/r32	LR	SGD	10^{-3}	1	0.05	–	0.678 MB	366.265 MB	177,804	78.79 ± 8.06	72.23 ± 7.91
SupCon 128/r32	LR	SGD	10^{-3}	1	0.1	–	0.678 MB	366.265 MB	177,804	78.66 ± 8.39	72.05 ± 9.07
SupCon 128/r32	LR	SGD	10^{-3}	1	0.01	–	0.678 MB	366.265 MB	177,804	78.53 ± 7.48	72.41 ± 7.45